Theme Article: Adversarial Examples in CE: Deceptive Threats to AI Systems

Deep Learning-Based Face Forgery Detection for Facial Payment Systems

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***Abstract*—The development of generative adversarial networks (GANs) has revolutionized image generation and editing. However, the capacity to create realistic images presents serious security concerns, particularly in the context of face-based payment systems. Deepfakes leverages GANs to generate manipulated videos or images, which may present opportunities for identity theft and fraudulent transactions. For instance, perpetrators employ Deepfakes technology to forge identifying information about victims, such as transplanting their faces into fake videos or images to make it appear like they are performing activities they have never done before. To address this growing concern, this study proposes a deep learning-based detection method utilizing an improved convolutional neural network model. The proposed model comprises two key modules, namely the multiscale attention (MA) module and the halo attention (HA) module. Specifically, MA is designed to recognize faces and other details in the forged image. HA is**

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built to focus on localized regions of the image. Experimental results show that the proposed model scores 97.12 and 99.32 on FF++ (HQ) dataset and 91.26 and 95.43 on FF++ (LQ) dataset in terms of accuracy and area under curve, respectively. The remarkable accuracy and performance make it a dependable solution for safeguarding face payment systems.

Facial payment systems have gained widespread adoption across various domains such as retail, finance, and public transportation. This innovative technology can enhance user experience and transaction efficiency. For instance, facial payments enable contactless transactions and minimize the risk of virus and bacteria transmission. Furthermore, eliminating physical cards through facial payments streamlines the payment process and enhances user convenience. However, the burgeoning field of deepfakes technology threatens the facial payment security.[1] It results in unauthorized transactions and the exposure of sensitive user information. Although some financial systems employ security measures, such as detecting vital signs like blinking or head movement, the continuous evolution of deep forgery techniques may breach these security measures.

Deepfakes is a technology that utilizes artificial intelligence techniques, e.g., generative adversarial networks (GANs)[2] to produce highly realistic fake videos or images. The working principle of GANs is illustrated in Figure 1. It comprises two modules, i.e., a generator that synthesizes new data from a target image or video, and a discriminator that attempts to distinguish the generated content from real data. Through an iterative refinement process, the generator progressively improves its forgeries to deceive the discriminator, ultimately producing highly convincing deepfakes. This poses a significant challenge to facial recognition systems, as it becomes increasingly difficult to differentiate between a real person and a deepfake.

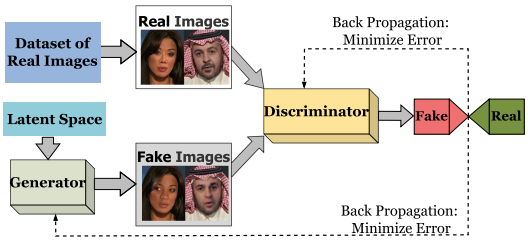

FIGURE 1. GANs network architecture.

To detect face forgery in consumer electronics, this work proposes a deep learning-based approach for forgery face detection. The overall framework of the proposed MAHA-Net is shown in Figure 2. Two modules are introduced in this network i.e., the MA module is designed to recognize faces and other details in the forged image. The halo attention (HA) module is built to focus on localized regions of the image, which helps to recognize subtle changes in expressions and movements of faces in forged images. The main contribution of this article is as follows:

- To improve the detection accuracy of the deepfake model, a deep leaning-based MAHA-Net is proposed by incorporating a MA module and a HA.
- A MA is introduced to capture forgery traces at various scales. Meanwhile, a halo self-attention model is adopted to capture useful relationships between nearby pixels.
- Comparative experiments conducted on public datasets verify that the proposed model outperforms the state-of-the-art methods in detection accuracy.

The remainder of this article is structured as follows. The "Related Work" section reviews existing methods used for deepfake detection. The "The Proposed Model" section delves into the details of the proposed deep learning-based method for deepfake detection. The "Experimental Results and Analysis" section describes the experimental setup, the datasets used, and the evaluation of the performance of the proposed method. Finally, the main conclusions of this work with some possible future trends are given in the "Conclusion and Future Work" section.

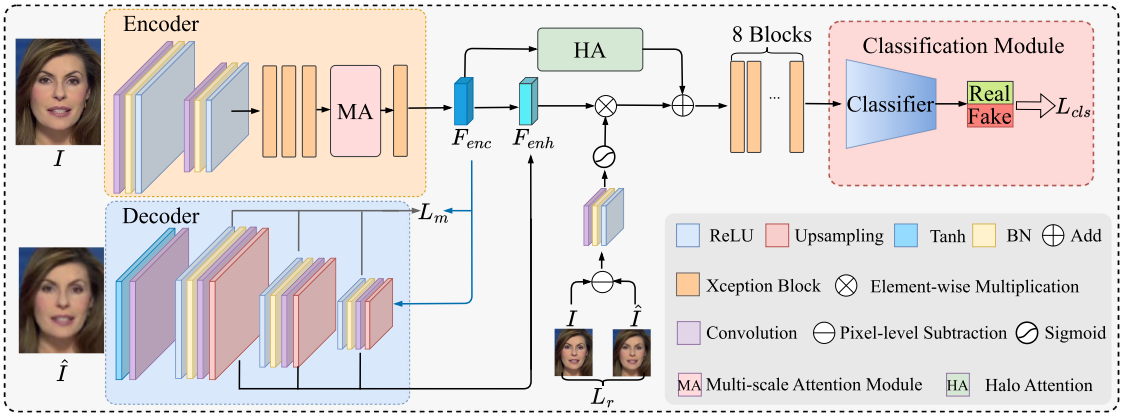

FIGURE 2. Overall framework of the proposed MAHA-Net.

RELATED WORK

Deep forgery detection is to identify and verify the authenticity of deeply forged images or videos, and researchers have given it considerable attention.

Early deepfake detection methods are mainly based on facial cues, e.g., head movements and facial expressions. Jung et al.[3] detected deepfakes by analyzing significant changes in the pattern of eye blinking. Ciftci et al.[4] leveraged biological signals as implicit authenticity descriptors to detect synthetic content in portrait videos.

Recently, deep learning has dominated the field of deepfake detection.[5,6] Zhou et al.[7] proposed a two-stream neural network for deepfake detection. Specifically, one branch is to analyze the visual appearance and the other focuses on local noise patterns. Rössler et al.[8] boosted the performance of deepfake detection by retraining an XceptionNet on manipulated face datasets. Li et al.[9] proposed a frequency-aware discriminative feature learning framework for face forgery detection. Nguyen et al.[10] proposed a capsule network to improve the deepfake detection accuracy. Cao et al.[11] proposed a reconstruction–classification learning (RECCE) framework, which learns differences in real face images through reconstruction.

PROPOSED MODEL

The proposed MAHA-Net framework is depicted in Figure 2. Compared with baseline RECCE,[11] the MA module and the HA module are employed in this network. The reconstruction network consists of an encoder and a decoder. Initialization of the encoder involves utilizing a pretrained Xception model.

Within the encoder, the MA module is positioned between the two block layers of the Xception. The input image I is initially processed by the encoder to extract feature maps, denoted by F_{enc}. Following this, F_{enc} is directed through two separate pathways. In one pathway, F_{enc} undergoes a multiscale graph reasoning operation within the decoder to obtain an enhanced feature representation, denoted by F_{enh}. Simultaneously, the reconstruction differences are utilized to guide the subsequent classification process. The other pathway involves applying the HA module to the feature layer F_{enc}. Finally, the feature maps obtained from both pathways are summed up and used as the basis for the classification module, which discriminates between real and fake content.

The output of the image after the reconstruction network is $\hat{I}$. The addition of noise enlarges the coding region of the image, thus masking out the distorted blank coding points. The reconfiguration network equation is as follows:

$$\hat{I} = f_{\text{rec}}(\tilde{I}) \tag{1}$$

where the variable $\tilde{I}$ denotes the output obtained by introducing white noise during the training period. $f_{\text{rec}}(\cdot)$ denotes the reconstruction network process.

MA Module

The MA module integrates both channel attention and spatial attention. Therefore, it

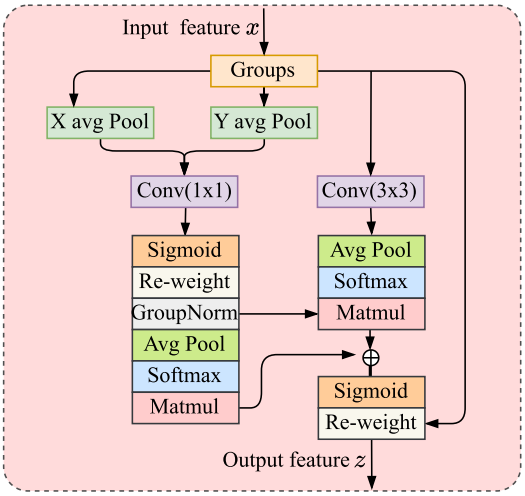

FIGURE 3. Schematic diagram of the MA module.

enables the simultaneous learning of interchannel and spatial feature dependencies. The architecture of the MA is shown in Figure 3. "Groups" indicates the divided groups, "X Avg Pool" means the 1D horizontal global pooling, and "Y Avg Pool" represents the 1D vertical global pooling, respectively. "Matmul" block is a matrix multiplication operation used to compute interactions between different features.

To acquire various semantic information, the MA module partitions the input image feature map x into g groups along the channel dimension. To capture features at various spatial scales, the MA module employs a three-parallel path structure. Two pathways utilize 1×1 convolutions to focus on fine-grained details within the local area of each pixel. The third pathway employs a 3×3 convolution to capture broader spatial relationships between pixels across a wider area.

HA Module

The HA[12] concentrates on a smaller localized window around each pixel. This module enables the proposed framework to capture useful relationships between nearby pixels. The architecture of the HA is shown in Figure 4.

Specifically, the input feature map is partitioned into four equally sized blocks. Each block is subjected to a padding operation, i.e., n layers of halos are added around each block. This augmentation expands the perceptual field of each block, and it results in a larger perceptual field

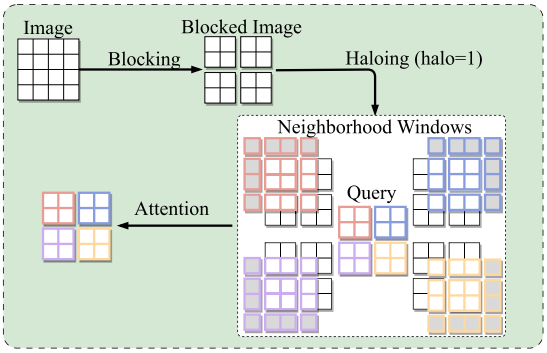

FIGURE 4. Schematic diagram of the HA module.

for each. Subsequently, each block is individually sampled. Ultimately, the feature map is output and combined with the initial input feature map. This seamless integration is facilitated by the utilization of the residual jump connection, which effectively amalgamates both local and global details.

Loss Function

The overall loss function is composed of three key components, namely reconstruction loss ($\mathcal{L}_{\mathrm{r}}$), classification loss ($\mathcal{L}_{\mathrm{cls}}$), and metric learning loss ($\mathcal{L}_{\mathrm{m}}$). It is formulated as

$$\mathcal{L} = w_1 \mathcal{L}_{\mathrm{m}} + w_2 \mathcal{L}_{\mathrm{r}} + \mathcal{L}_{\mathrm{cls}} \tag{2}$$

where the metric learning loss $\mathcal{L}_{\mathrm{m}}$ minimizes the distance between real data and maximizes the separation between real and synthetic representations in the embedding space. Following the setup described in RECCE,[11] the values of w_1 and w_2 are set to 0.1.

EXPERIMENTAL RESULTS AND ANALYSIS

This section details the dataset used, the experimental setup employed, and the analysis of the obtained results. Comparative experiments were conducted on FaceForensics++ (FF++)[8] dataset. FF++ dataset is a forensics dataset comprising 1000 original video sequences manipulated with four automated face manipulation methods. Each video is processed using one of four state-of-the-art face synthesis methods, namely Deepfakes, Face2Face, FaceSwap, and NeuralTextures. To evaluate the performance of the proposed method, we used two metrics,

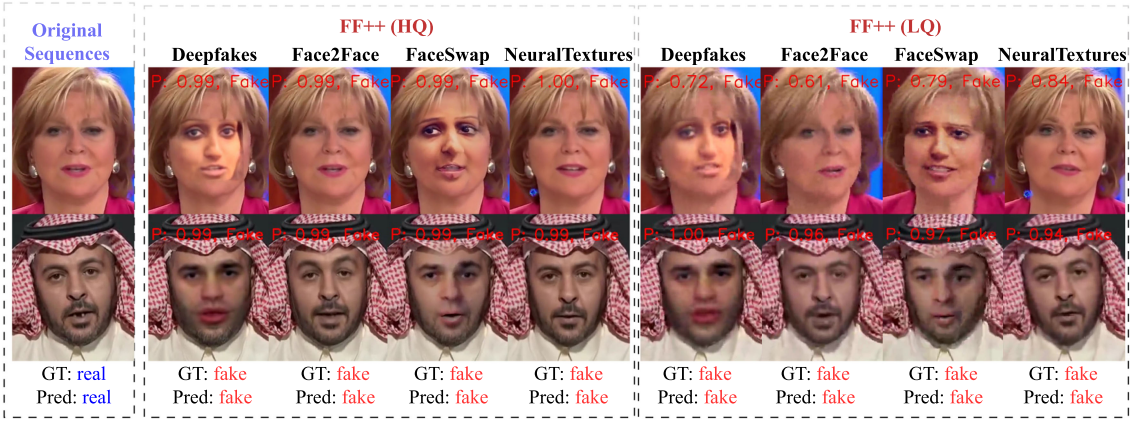

FIGURE 5. The inference results of deepfake detection by the proposed model.

namely accuracy (ACC) and area under the curve (AUC). The ACC is the ratio of correctly predicted samples to the total number of samples. It provides a straightforward measurement of the model's ability to classify both real and fake images. However, ACC can be misleading with class imbalance. The AUC is a more robust metric that is not affected by class imbalance. Higher ACC and AUC indicate better discrimination ability of the model. We trained the model with 32 samples per batch for 40 iterations. The learning rate is dynamically adjusted during training using a step-learning rate scheduler. We implemented the framework using PyTorch[13] and trained it on two 3090 Ti GPUs in parallel.

As shown in Table 1, the proposed method performs best on FF++ datasets. Compared to the baseline RECCE,[11] the proposed method achieves the same excellent results on FF++ (HQ) dataset. On FF++ (LQ) dataset, the proposed model improves the ACC and AUC to 91.26% and 95.43%, respectively. These results indicate that the proposed network has an advantage in accurately detecting face forgery images compared with the SOTA competitors. The performance improvement can be attributed to the proposed MA and HA modules. Specifically, the MA module captures and integrates features across multiple scales, which enhances the ability of the network to detect subtle and varied patterns in deepfake images. The HA module focuses on the most relevant regions. It enables the model to distinguish between authentic and manipulated content and thus boost performance, especially in low-quality datasets. Low-quality data are susceptible to adversarial attacks.[20] This is because the noise and artifacts present in low-quality data can provide a hiding place for adversarial perturbations. Compared to the SOTA methods, the superior performance of the proposed framework on low-quality datasets verifies the robustness of the proposed model.

TABLE 1. Comparison with the SOTA methods. The best results are highlighted in bold.

Methods	FF++ (HQ)		FF++ (LQ)	
	ACC	AUC	ACC	AUC
SPSL[14]	91.50	95.32	81.57	82.82
RFM[15]	95.69	98.79	87.06	89.83
Multitask[16]	85.65	85.43	81.30	75.59
Xception[17]	95.73	96.30	86.86	89.30
Add-Net[18]	96.78	97.74	87.50	91.01
Two-branch[19]	96.43	98.70	86.34	86.59
Freq-SCL[9]	96.69	99.28	89.00	92.39
RECCE[11]	97.06	**99.32**	91.03	95.02
MAHA-Net (Ours)	**97.12**	**99.32**	**91.26**	**95.43**

The objective result of deepfake detection by the proposed model is shown in Figure 5. It shows that the proposed MAHA-Net can accurately predict the authenticity of human faces.

CONCLUSION AND FUTURE WORK

Deep face forgery poses a significant security risk to face payment systems in the consumer electronics domain. To mitigate these

risks, a deep forgery detection framework named MAHA-Net is presented in this work. MAHA-Net integrates a MA module and a HA module. The MA module captures image features at different scales, and the HA module focuses on localized regions of the image. The two modules can extract the forged features in deep forged images. Extensive experiments have demonstrated that MAHA-Net outperforms current state-of-the-art methods in accuracy on the widely used FF++ dataset.

The results indicate that the detection accuracy of the MAHA-Net and existing detection methods was low on the low-quality FF++ datasets. In the future, we will aim to improve the accuracy of low-pixel forged faces. In addition, we plan to investigate the integration of techniques like transfer learning to further enhance the robustness of MAHA-Net against new and unseen deepfakes.

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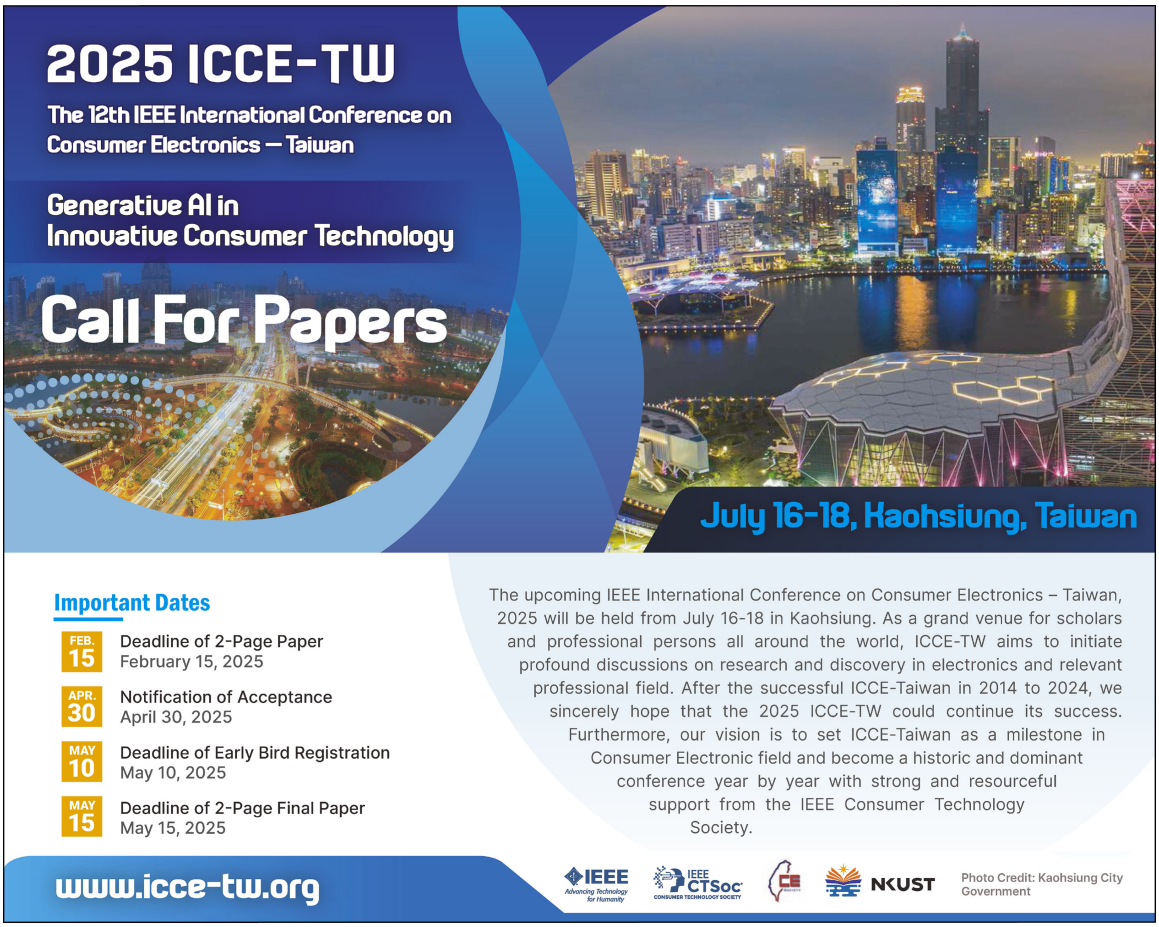
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